RESEARCH ROADMAP

From 28 methods to a real prediction test

This document uses simple technical English (ASD-STE100 style, about 80%). It states one idea per page. It uses facts from the v001 experiments.

1. What this document is

Core research question (docs/DATA.md): Which dependence measure is most useful for a later up, down, or crash read on these eight major FX pairs?

A list of 28 correlations is a catalogue. It is not the answer. A fair answer needs three parts:

- **A target** — what you try to predict (vol spike, drawdown, or correlation break).
- **A time-t feature** — a measure that uses only data available at day t.
- **An out-of-sample test** — train on the past, score on later data.

How to use the two PDFs

- **Correlation_Methods_28_Pages.pdf** — how each method works (the catalogue).
- **Research_Roadmap.pdf** (this file) — what to fix, what to test, and in what order.

Do not send the catalogue alone to a desk and call it a prediction paper. Close the gaps on the next pages first.

Page map of this roadmap

2 — Where the project is (assets and blockers)
3 — Seven gaps in one table
4–5 — Gap cards with done-conditions
6 — Known results vs your real edge
7–8 — Crash prediction honesty and how to read turbulence
9–10 — Eight research steps mapped to this repo
11 — Work board checklist
12 — What success is, and five desk questions

2. Where the project is now

Data: eight majors (EURUSD, GBPUSD, AUDUSD, NZDUSD, USDJPY, USDCHF, USDCAD, EURGBP). Timeframe D1. Snapshot `2026-11-03_majors_alpari_D1`. Range about 1999–2026 (~7,200 bars per pair).

Assets you already have

- A frozen price snapshot that all methods share.
- Twenty-eight implemented formulas with v001 indexes and pair reports.
- A watchlist of rolling breaks (method 05).
- System PCA and connectedness for the eight-major panel (method 28).
- A one-page method catalogue PDF for every formula.

Blockers you do not have yet

- Out-of-sample forecast scores versus Pearson and rolling Pearson.
- Bias and multiplicity controls on several methods.
- A written target definition for crash or stress labels.
- An early-warning test on 2008, SNB 2011–15, 2020, and 2022.

All current results describe the full sample. They do not yet forecast the next move. They do not yet give an out-of-sample early warning.

3. Seven gaps in the current results

Fix these problems before you claim prediction power. Each row is one gap from the v001 notes and cross-checks.

Gap	v001 fact	Fix
Cointegration p	Raw p<0.05 on a few pairs; 28 tests.	Bonferroni / FDR. Report corrected hits only.
Transfer entropy	Scores ≈ 0.004–0.009 nats; bias ≈ 0.0056.	Shuffle / surrogate tests before claims.
ADCC / DCC fits	EURUSD-USDCHF has b=0.5; some g=0 on bound.	Re-fit. Check convergence. LR-test g.
EURGBP in panel	PC8 eigenvalue ≈ 0.0046 (near singular).	Drop or orthogonalise for PCA / VAR.
Stress correlation	Raw stress rho rises with volatility.	Forbes–Rigobon adjust or bootstrap the gap.
Tail benchmark	Independence is the wrong yardstick.	Compare L_hat to Gaussian with same r.
No OOS test	All v001 work is full-sample description.	Build time-t features and forecast targets.

The next two pages open the main gaps as cards. Each card states what is wrong, the number, what you do next, and when the gap is closed.

4. Gap cards — cointegration and transfer entropy

Gap A1 — Cointegration p-values (method 27)

What is wrong	A few pairs show raw p below 0.05 (for example EURUSD-USDCAD). With 28 tests, about 1.4 false hits are expected at 5% by chance.
The number	Bonferroni cut = 0.05 / 28 ≈ 0.0018. After that cut, no pair stays significant in v001.
What you do next	Apply Bonferroni or FDR. State the honest finding: no robust cointegration on this design.
Done when	A v002 NOTES.md table shows raw p, corrected p, and which pairs survive.

Gap A2 — Transfer entropy bias (method 17)

What is wrong	Discrete plug-in TE with B=5 bins and lag 1 has a positive bias under independence.
The number	Bias size ≈ (B–1)^2·B / (2N) ≈ 0.0056 nats for N≈7200. Many TE scores sit between about 0.004 and 0.009 nats.
What you do next	Run shuffle or surrogate tests. Keep only scores that clear the null.
Done when	A v002 NOTES.md ranks directed TE only after the null test, with a clear pass/fail flag.

5. Gap cards — fits, EURGBP, stress, tails

Gap B1 — ADCC / DCC convergence (methods 09–10)

What is wrong	EURUSD-USDCHF shows b = 0.500000 while other pairs have b near 0.89–0.99.
The number	Some pairs also show g = 0.000000 on the bound.
What you do next	Re-fit. Check convergence. Run a likelihood-ratio test for g.
Done when	Every pair has a status flag and a documented fit check in v002.

Gap B2 — EURGBP redundancy (method 28)

What is wrong	EURGBP is close to EURUSD minus GBPUSD in log returns.
The number	PC8 eigenvalue ≈ 0.0046 — the panel is almost singular.
What you do next	Drop EURGBP from system VAR/PCA, or orthogonalise it.
Done when	System metrics rerun without the redundant leg, with NOTES updated.

Gap B3 — Stress correlation bias (method 11)

What is wrong	Raw correlation often rises when volatility rises, even if structure is unchanged.
The number	Largest |stress – quiet| gaps include AUDUSD-USDJPY ≈ +0.267.
What you do next	Apply Forbes–Rigobon adjustment or bootstrap the gap.
Done when	Quiet/stress table reports adjusted gaps with a significance note.

Gap B4 — Tail benchmark (method 19)

What is wrong	Comparing L_hat only to independence is the wrong yardstick.
The number	Example: AUDUSD-NZDUSD lower-tail L_hat ≈ 0.583 at alpha=0.05.
What you do next	Compare L_hat to the co-exceedance a Gaussian with the same Pearson r would give.
Done when	Each pair has L_hat, Gaussian benchmark, and excess = L_hat – benchmark.

6. Results that are real but already known

Keep these as validation. Do not sell them as the only discovery. A desk will ask: what is new after these known facts?

Dollar factor

Method 07: EURUSD–AUDUSD Pearson r ≈ 0.557 falls to rho_partial ≈ –0.016 after the USD control. Much of the raw link was the dollar.

Carry crash risk

Method 19: AUDUSD–USDJPY has near-zero full-sample r (≈ –0.088) but lower-tail L_hat ≈ 0.169. Crash co-movement is larger than the average link.

Asymmetric correlation

Method 10: USDCHF–USDCAD has the largest g ≈ 0.047. Joint downs can move correlation more than joint ups.

Connectedness

Method 28: system TCI ≈ 54.0; EURUSD net ≈ +21.7%. EURUSD acts as a net shock transmitter in this panel.

Your edge: the joint bank of 28 methods on one frozen snapshot. Use that edge for a forecast horse race, not for a longer list of static tables.

7. Can we predict a crash like 2008 or COVID-19?

No method here gives the exact date of a crash. Markets do not publish a calendar of crises. Do not promise date prediction to institutes or desks.

Predict the date	**Detect risk build-up**
Name the day of a crash in advance. **Do not claim this.**	Show that selected measures move before stress windows more often than Pearson alone. **This is the paper.**

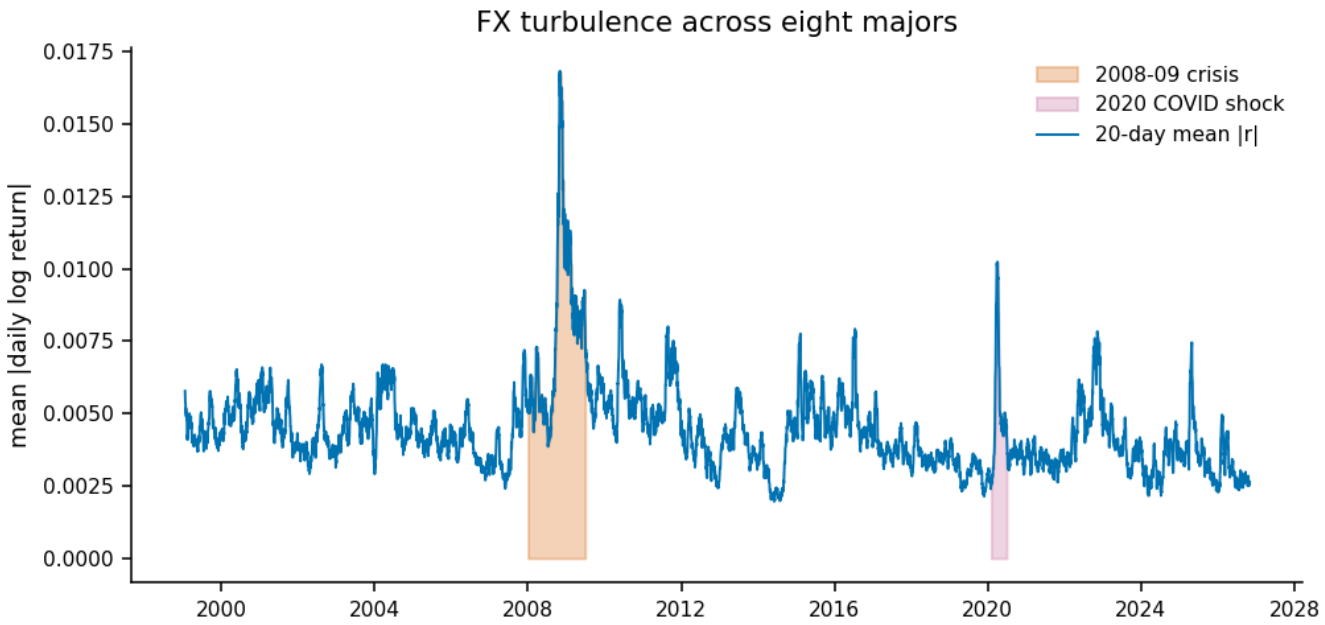


Figure: 20-day mean of equal-weight |daily log returns| across the eight majors. Orange shade = 2008–09. Pink shade = early 2020. Turbulence rises in those windows.

8. How to read the turbulence chart — and what to test

What the axis means

Each day, take the absolute log return of each of the eight majors. Average those eight values. Then take a 20-day mean of that series. A high value means the majors moved a lot together in absolute terms.

What the shaded bands are

- **2008–09:** global financial crisis window.
- **Early 2020:** COVID market shock window.

The research test (write this in NOTES before you run it)

At each day t before a shaded window, compute features with data ≤ t only (for example ADCC – Pearson, or |r_20 – baseline|). Ask: do these gaps rise in the weeks before the shade, or only inside the shade?

Score hit rate and false-alarm rate against rolling Pearson. A signal that fires only after the crash has low value for desks.

Crisis windows to mark

- 2008–09 global financial crisis
- SNB EUR/CHF floor period and Jan 2015 removal
- 2020 COVID shock
- 2022 rate-shock / risk-off period

9. How to research the core issue — eight steps

Do the work in this order. Do not jump to a custom formula before the tests.

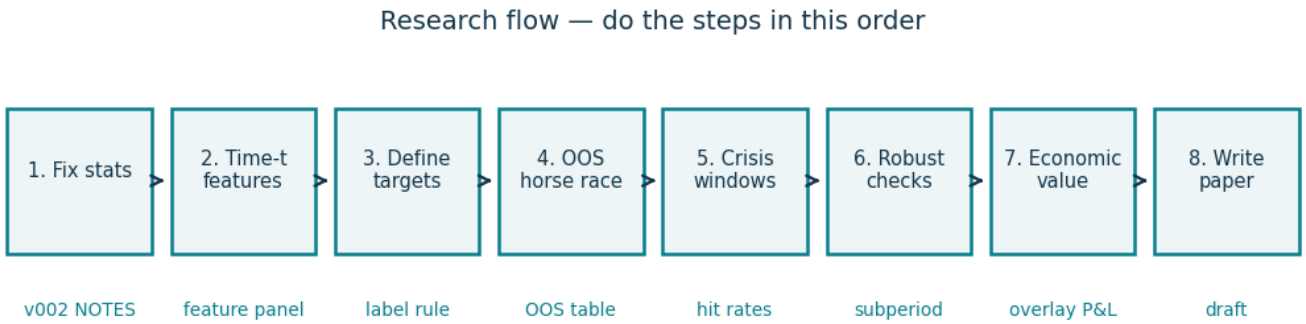


Figure: each box is a step. The label under the box is the file or table you must produce.

Step 1 — Fix the statistical problems

Do this: Close gaps A1–B4 (multiplicity, TE null, DCC fits, EURGBP, stress bias, Gaussian tails).
Produce this: v002 NOTES.md in each affected method folder.
Done when: Every gap has a corrected table and a one-line done statement.

Step 2 — Build time-t features

Do this: Expanding or rolling windows with no future data for the shortlist.
Produce this: A dated feature panel for methods 01, 05, 07, 10, 11/19, 14, 28.
Done when: Every cell uses only information available at date t.

10. Steps 3–8 and how they map to this repo

Step 3 — Define the targets

Do this: Pick one or two labels before you look at results.
Produce this: A written label rule in NOTES.md.
Done when: The rule is frozen before any forecast score is computed.

Example targets (choose before you look)

- Next-20-day realised-vol spike above the 90th percentile of past vol.
- Pair drawdown beyond a fixed threshold in the next 20 days.
- Correlation break: |r_20 − baseline| above a pre-set cut.

Step 4 — Out-of-sample horse race

Do this: Train on past data. Score on later data. Compare each measure to Pearson and rolling Pearson.
Produce this: OOS score table with Diebold–Mariano tests and a model confidence set.
Done when: A clear winner set — or a clear statement that nothing beats the baseline.

Step 5–8 — Crisis tests, robustness, economic value, paper

Do this: Run early-warning on crisis windows; repeat on subperiods; test a simple overlay with costs; write the paper.
Produce this: Hit rates, subperiod tables, overlay P&L, draft with the 28-method PDF as appendix.
Done when: You can answer the five desk questions on page 12 with tables.

Version rule

New work is experiments/v002, v003, … Do not overwrite v001. See docs/VERSIONING.md.

Shortlist for deep work

01 Pearson · 05 Rolling · 07 Partial · 10 ADCC · 11 Regime · 19 Tail · 14 Distance correlation · 28 Networks / PCA / Connectedness. Keep the other methods in the appendix catalogue.

11. Work board — checklist and order

Use this table as your work board. Finish row 1 before you invent a new formula.

#	Task	Why	Output
1	Fix statistical gaps	Stops false claims before forecasts.	Method folders: experiments/v002/NOTES.md
2	Build time-t measures	No look-ahead in features.	Dated feature panel (CSV/parquet)
3	Define targets	Makes the question testable.	Written label rule in NOTES.md first
4	OOS horse race	Shows who beats Pearson.	OOS score table + DM / MCS
5	Crisis window tests	Checks 2008, SNB, 2020, 2022.	Hit / false-alarm rates by window
6	Robustness	Result must hold in more than one decade.	Subperiod tables
7	Economic value	Links stats to desk use.	Simple hedge / vol overlay with costs
8	Write the paper	One clear claim for institutes and desks.	Draft; 28-method PDF as appendix

Daily habit

- Open one gap. Close it. Write a short NOTES.md. Stop.
- Do not mix a new formula and a new forecast test in the same day.
- Keep v001 folders immutable. Always create v002, v003, …

12. What success is — and what a desk will ask

Success is not “28 correlations exist.” Success is a clear, tested statement such as:

> “On this frozen majors panel, measure X improves next-horizon stress or correlation-break forecasts versus rolling Pearson, out of sample, after multiplicity control.”

If no measure beats the baseline, that result is still publishable. Report it without force.

Five yes/no questions a hedge fund will ask

1. Did you use only time-t information?
2. Did you beat a simple rolling Pearson baseline?
3. Did you control for many tests?
4. Did the signal work before 2008/2020, not only inside the crash?
5. Is there a simple economic use case with costs?

How to rebuild this PDF

From the repo root:

```
python reports/build_research_roadmap.py
```

Method catalogue:

```
python reports/build_methods_overview.py
```

Prepared for CU Research One — Correlation. Writing aims for ASD-STE100 style (about 80%). Formula names and statistical terms are exempt.